Table 5: Hyperparameter tuning results for ASSIST2009

Corrected causal LightSAKT · validation results

Learning Rate	Embedding	Heads	Dropout	Best validation AUC
0.000500	32	2	0.1	0.743801
0.000500	32	2	0.2	0.743286
0.000500	32	4	0.1	0.744715
0.000500	32	4	0.2	0.744041
0.000500	64	2	0.1	0.750194
0.000500	64	2	0.2	0.750272
0.000500	64	4	0.1	0.749702
0.000500	64	4	0.2	0.749725
0.000500	128	2	0.1	0.750494
0.000500	128	2	0.2	0.750732
0.000500	128	4	0.1	0.750508
0.000500	128	4	0.2	0.750191
0.001000	32	2	0.1	0.745476
0.001000	32	2	0.2	0.744907
0.001000	32	4	0.1	0.745685
0.001000	32	4	0.2	0.745398
0.001000	**64**	**2**	**0.1**	**0.750804**
0.001000	64	2	0.2	0.750408
0.001000	64	4	0.1	0.750640
0.001000	64	4	0.2	0.750068
0.001000	128	2	0.1	0.749276
0.001000	128	2	0.2	0.750502
0.001000	128	4	0.1	0.750434
0.001000	128	4	0.2	0.750732
0.005000	32	2	0.1	0.747109
0.005000	32	2	0.2	0.746947
0.005000	32	4	0.1	0.745964
0.005000	32	4	0.2	0.745198
0.005000	64	2	0.1	0.748094
0.005000	64	2	0.2	0.748185
0.005000	64	4	0.1	0.749193
0.005000	64	4	0.2	0.748850
0.005000	128	2	0.1	0.748437
0.005000	128	2	0.2	0.748855
0.005000	128	4	0.1	0.748005
0.005000	128	4	0.2	0.748094

LightSAKT; all 36 configurations. Training seed 42, split seed 42; validation AUC only.
Bold marks the highest full-precision validation AUC. Exact ties select the lexically smallest run ID.
Selected: LR 0.001, dimension 64, heads 2, dropout 0.1; epoch 24.